VIDEO MANAGEMENT & RESTREAMING PLATFORM

One platform for every camera, every stream, every stakeholder.

XFlow lets transportation agencies view, control, secure, and share live camera video — from the operations center to the traveling public — on a single, highly scalable platform.

5,000+	as low as 250 ms	ONVIF / NTCIP	REST API
cameras per deployment	ultra-low latency	camera control	for every capability

THE PLATFORM

What XFlow is

The first unified, cost-effective solution for both video management and scalable restreaming in the TSMO domain.

XFlow is a feature-rich platform for camera control and management, video normalization, and secure restreaming — with intelligent real-time monitoring, a modern browser-based interface, and a fully featured REST API for external-system integration. It scales through the simple addition of XEngine instances, and deploys on private virtualized environments, bare-metal servers, or commercial cloud.

BUILT FOR THE MANY WAYS AGENCIES PUT VIDEO TO WORK:

- Traffic operations & video walls
- 511 / traveler-information web & mobile apps
- GIS departments (e.g. ArcGIS)
- Video analytics & machine learning
- Emergency management
- Partner agencies
- First responders
- Media organizations

CAPABILITIES

A comprehensive platform, end to end

VIEWING & CAMERA CONTROL

Best-in-class viewing & control

Fast, intuitive live viewing and responsive PTZ control, with stream tours and camera preset tours. Work from whichever view suits the task — a filterable List with live status and snapshots, the Camera map, or Personal wall.

Camera map

Every camera on a map, colored live by whether its stream and PTZ are available. Filter by status, search by name, and drag a camera onto the bordering video panel to watch it without leaving the map. An optional self-hosted basemap covers sites with no internet.

Recording, playback & video library

Record streams continuously, via schedule, or on-demand, with configurable storage and retention, for secure browsing and playback — and upload MP4 files, such as training videos, for playback by authorized users under the same token protection.

Automated stream configuration

Define encoding templates once; XFlow provisions matching profiles across your ONVIF cameras, verifies them every polling cycle, and repairs drift — down to vendor quirks such as Bosch GOP structure.

Single sign-on

Sign in through Microsoft Entra or any OpenID Connect provider, inheriting your organization's MFA and password policy. Provider-managed and local accounts coexist on one deployment, with roles and permissions owned by XFlow.

Personal wall

Every operator can build their own wall in the browser — fixed grids, or a flex layout that subdivides any cell into quads — assigning streams and tours by drag and drop. Cameras can be controlled in place.

Protocol-agnostic camera control

Control cameras over ONVIF or NTCIP 1205 (SNMP), selectable on a camera-by-camera basis.

Video-wall integration

Read the layouts from XCube, Datapath, and Barco wall controllers so any browser can mirror control-room walls.

STREAMING & DELIVERY

Broad, modern format support

Restream to browsers, mobile devices, workstations, video walls, recorders, and external systems.

On-demand ingestion

Ingest an input stream only while it is being viewed — ideal for metered / wireless cameras and resource-constrained hosts.

Ultra-low latency

As low as 250 ms for real-time applications such as camera control and incident detection — delivered over WebRTC.

Input conversion

Re-multiplex or transcode on the fly: support legacy formats and adapt bitrate, resolution, and frame rate before restreaming.

SCALE & DISTRIBUTION

Multicasting & publishing

Efficient one-to-many RTSP multicast, plus stream 'push' to partner servers over RTSP/S and RTMP/S.

Composite streams

Combine multiple live streams into a single composite stream that is viewed, delivered, and recorded like any other.

Single-hostname stream directory

Publish every stream under one hostname: viewers are redirected to the serving engine — or handed a LAN multicast join — without wasteful proxying.

CDN overflow delivery

Optionally deliver HLS / LLHLS through a CDN edge for very large public audiences. Opt in per protocol and per server: viewers are handed the edge URL rather than the engine's.

SECURITY & MONITORING

Multi-layered stream security

Request-header validation, rotating token authorization, automatic blacklisting by address or ASN, and host-firewall IP white/blacklisting — deterring commercial piracy of public streams without impacting viewers.

Active monitoring

Real-time status and performance for servers, streams, and cameras, with a rolling history exposed through the REST API. Streams report codec, resolution, bitrate, and client count; hosts, CPU, memory, and disk.

Connection monitoring & audit history

Live client-connection visibility, plus audit trails for logins, stream requests, blacklisting, and camera control. Administrator access can be limited to listed networks, with per-account session lifetimes and remote sign-out.

Certificate rotation without downtime

Install a renewed TLS certificate over the REST API — no restart, no dropped viewers — and push it to a co-deployed XSnaps.

Edge servers

Front the public with an engine that has no route to the camera network: edge streams pair with their origins automatically, each viewer sees the half their network belongs on, and wall presets and tours play from either side.

Load balancing & failover

Distribute load across XEngine instances with economical single-tier balancing and host-failover support.

INTEGRATION & EXTENSIBILITY

Open REST API

Every platform capability — configuration, monitoring, and low-latency camera control — available to external systems. Configuration also round-trips through Excel for bulk editing.

Analytics / AI integration

Feed snapshots, live streams, or pushed frame-clips to video-analytics and ML systems, minimizing bandwidth.

Audience-scoped stream suspension

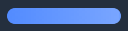
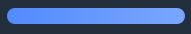
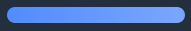
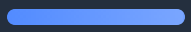
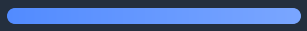
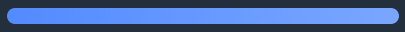
Suspend or release named streams for a stream group over the API, withholding a camera from the public while operators keep watching.

Snapshot publishing

Generate and archive still images on a schedule — with logos, timestamps, and thumbnails — to folders, network shares, or FTP.

INTEROPERABILITY

Protocols & latency

OUTPUT PROTOCOL	RELATIVE LATENCY	LATENCY	TYPICAL USAGE	CODECS
WebRTC / WHEP		< 500 ms	Web client — camera control	H.264 H.265
RTSP / UDP		< 1 s	Native client or server	H.264 H.265
SRT		< 1 s	Native client or server	H.264 H.265
RTSP / TCP		< 2 s	Native client or server	H.264 H.265
RTSPS		< 2 s	Native client or server	H.264 H.265
RTMP / RTMPS		1–2 s	Native client or server	H.264 H.265
LLHLS		4–10 s	Web client — mass distribution	H.264 H.265
HLS		12–30 s	Web client — mass distribution	H.264 H.265

SUPPORTED INPUTS

RTSP / UDP (RTP, RTCP) · RTSP Multicast · RTSP / TCP · RTSPS · RTMP · RTMPS · Multicast TS · HLS · LLHLS

Non-exhaustive. Additional protocols and codecs supported via re-multiplexing and transcoding. H.264 and H.265 (HEVC) inputs are both supported; WebRTC/WHEP is not offered for H.265, due to web browser (decoding) limitations.

ARCHITECTURE & DEPLOYMENT

Built to scale, deployed your way

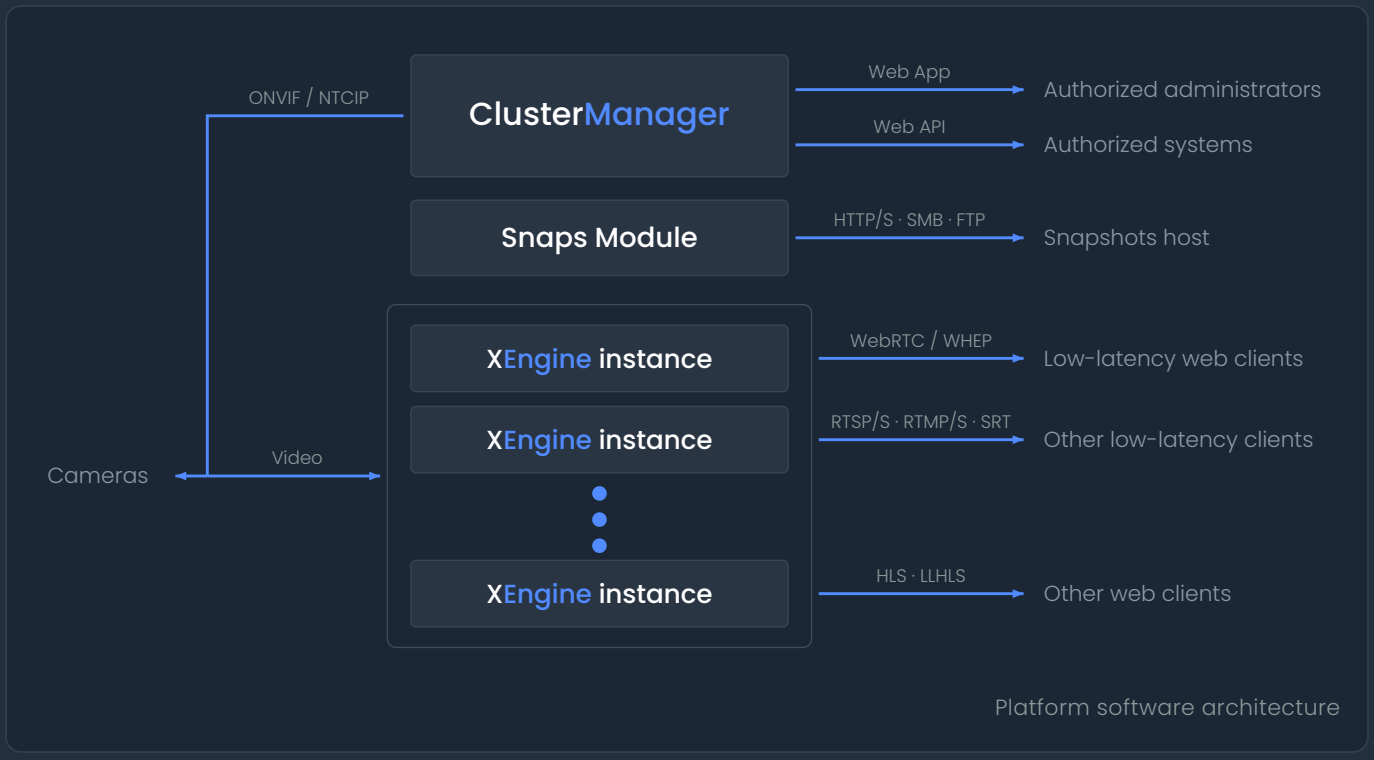
XEngine Instance

The XFlow Restreaming Engine — Go and C/C++ — delivers high-quality restreaming at low latency and resource cost.

XFlow Cluster Manager

The central component: control, monitoring, and configuration for the entire platform.

Scale horizontally by adding XEngine instances. Deploy on private virtualized environments, bare-metal servers, or commercial cloud.



Platform software architecture

Proven in production on statewide traveler-information systems, serving live video to the public at scale.

Bring XFlow to your operation.

Additional protocols and codecs are available via re-multiplexing and transcoding. Contact Optima DTS to discuss your deployment or arrange a demonstration.

contact@optima-dts.com · optima-dts.com